

EchoGuard: Continuous Ultrasound-based Human Presence Verification Against AI Agents

Yizhu Wen^{†*}, Dharani Nadendla^{‡*}, Shirui Cao[§], Renzhi Hao[¶], Dong Li[‡], Hanqing Guo[†]

[†]Indiana University Bloomington [‡]University of Maryland, Baltimore County

[§]University of Massachusetts Amherst [¶]Independent Researcher

{yizwen, guohan}@iu.edu

{dharann1, dli}@umbc.edu shiruicao@cs.umass.edu horange1024@outlook.com

Abstract—Autonomous web AI agents are increasingly capable of imitating human interaction, challenging existing defenses that rely on point-in-time or software-based checks. We present EchoGuard, a browser-based prototype that explores the feasibility of continuous, action-coupled human-presence verification using active ultrasound. EchoGuard uses laptops’ built-in speakers and microphones to transmit dual-band triangular frequency-modulated continuous-wave (FMCW) signals and align the received acoustic responses with browser input events. We evaluate EchoGuard on four commodity laptops under typing, touchpad interaction, surrounding human-motion interference, and software-generated agent interaction. Our results show that Doppler provides the clearest event-aligned representation, direct interactions exhibit patterns distinct from surrounding motion, and event-level Doppler features reveal distinct human- and agent-generated interaction distributions, enabling three binary classifiers to distinguish typing events with accuracies up to 89.6%. These findings provide initial evidence that active ultrasound can support future continuous human-presence verification against AI agents.

I. INTRODUCTION

Recent advances in large language models (LLMs) and multimodal perception have made autonomous web AI agents increasingly capable of operating on realistic and live websites [1], [2], [3]. Modern agents, such as *OpenClaw* [4], can perform account registration, checkout automation, and data scraping with minimal human involvement. Unlike traditional bots, these agents can adapt to interface changes and imitate realistic user behavior, making them difficult to distinguish from legitimate users [5], [6]. This creates a growing security challenge: distinguishing security-sensitive actions performed by humans from those performed by AI agents.

Existing defenses provide only limited protection. CAPTCHA [7] and Multi-Factor Authentication (MFA) [8] are typically one-time checks and can be bypassed through relay services or human-solving farms [9], [10], [11]. Device fingerprinting and behavioral detection are vulnerable to emulation and adaptive agents that mimic user interactions [12], [13]. More fundamentally, these mechanisms do not continuously verify that a human remains physically involved during an active session [14]. Once authentication succeeds, an AI agent can generate keyboard, pointer, and scrolling actions in software, producing browser events that closely resemble those triggered by a legitimate user.

Although an AI agent can synthesize browser events, it does not by itself produce the physical finger and hand movements associated with typing on a keyboard or operating a touchpad. This motivates us to investigate physical sensing as a complementary signal for human-presence verification. Commodity laptops already contain speakers and microphones capable of emitting ultrasonic signals and sensing fine-grained human motion from their reflections [15], [16]. Prior acoustic sensing systems have demonstrated gesture and touch sensing. However, it remains unclear whether commodity laptop acoustics can capture physical interaction signals that are sufficiently associated with browser events to distinguish human interaction from software-generated actions.

To overcome this challenge, we present EchoGuard, a browser-based prototype that explores the feasibility of continuous, action-coupled human-presence verification using active ultrasound. EchoGuard uses a laptop’s built-in speakers and microphone to transmit dual-band triangular frequency-modulated continuous-wave (FMCW) signals [17] and capture reflections associated with human interaction. It aligns amplitude, phase, and micro-Doppler responses with events logged by the browser. Before transmitting the sensing signal, it filters audible frequencies locally to reduce privacy risks.

We conduct a feasibility study on four laptops with different speaker-microphone configurations. Specifically, we characterize their acoustic responses during typing and touchpad interactions, in the presence of nearby human motion, and during software-generated agent interactions. Our measurements yield several findings: (i) Doppler provides the clearest event-aligned representation among the evaluated acoustic features; (ii) direct keyboard and touchpad interactions produce acoustic patterns that are distinct from those caused by surrounding human motion, although nearby motion can introduce interference; (iii) the same sensing pipeline captures interaction-related responses across different laptop configurations; and (iv) event-level Doppler features exhibit distinct distributions for human- and agent-generated interactions, enabling three binary classifiers to distinguish human from agent-generated typing with accuracies around 88.0%, a macro F1 and an AUC up to 0.793 and 0.759, respectively. These results provide initial evidence that active ultrasound can serve as a physical signal for continuous human-presence verification. Our contributions can be summarized as follows:

*Yizhu Wen and Dharani Nadendla contributed equally to this work.

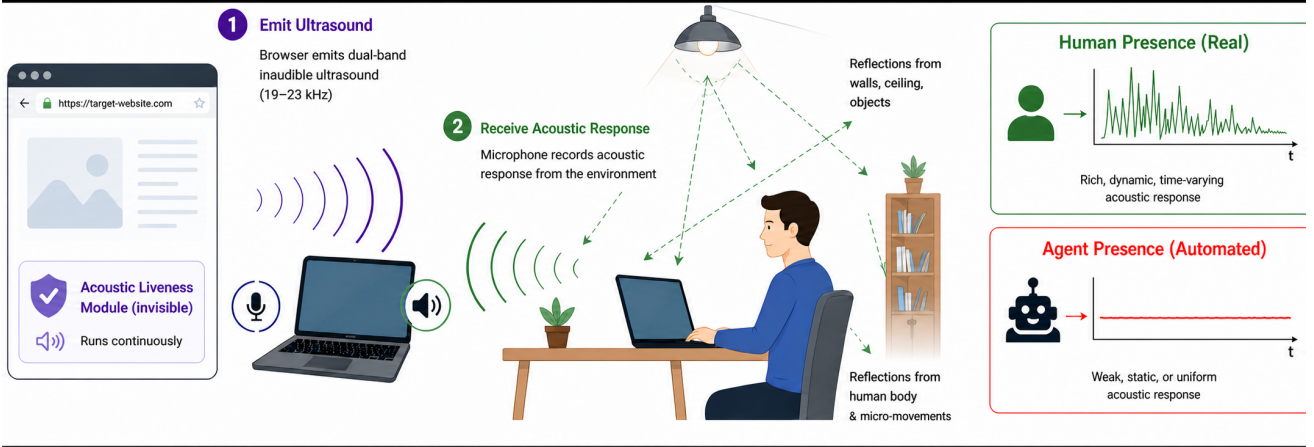


Fig. 1: EchoGuard deployment scenario. An EchoGuard-protected website uses the client’s built-in speakers and microphone to transmit ultrasonic signals, capture their reflections, and examine whether security-sensitive actions are performed by a human.

- We explore action-coupled physical ultrasound sensing as a complementary signal for continuously verifying human presence against web AI agents.
- We develop EchoGuard, a browser-based active-ultrasound prototype that captures and processes interaction-related acoustic reflections on laptops.
- We conduct a feasibility study across four laptops and multiple interaction conditions, characterizing how Doppler responses vary with human interaction, surrounding motion, and software-generated agent activity.

II. RELATED WORK

A. Bot Detection

Web services use CAPTCHAs [7], MFA [8], browser fingerprinting [18], [19], and behavioral analysis [20], [21] to distinguish legitimate users from automated clients. CAPTCHAs present explicit human challenges, but they can be bypassed through commercial solving services or automated techniques [9], [10]. Browser fingerprinting identifies clients from software, hardware, and rendering features. However, these features can be spoofed or reproduced in controlled automation environments [12], [13].

B. Acoustic Sensing and Liveness Detection

Commodity speakers and microphones have been widely used for human activity sensing. Previous systems [22], [15], [23], [24], [25], [26] detect gestures, finger motion, or touch interactions from acoustic signals. Acoustic sensing has also been used for liveness detection: VoiceLive [27] detects live speech, while VoiceGesture [28] verifies articulatory motion using Doppler shifts. In contrast, EchoGuard does not recognize gestures or verify speech liveness. It investigates whether browser events are accompanied by corresponding physical interaction with the client device, using ultrasound as a complementary signal for continuous human-presence verification.

III. THREAT MODEL

We consider an *attacker who uses autonomous AI agents to interact with websites with minimal human intervention*. Such

agents may scrape protected content, abuse promotions, or automate actions on compromised accounts while interacting through standard browser interfaces and imitating legitimate user behavior. The *defender is the website owner*, whose goal is to identify signals that can help distinguish autonomous-agent interactions from legitimate human activity without substantially disrupting usability.

Attacker capability: We assume that the attacker can use AI agents capable of generating browser interactions, including keyboard, pointer, scrolling, and clicking events. The attacker may also adapt the agent’s software behavior to resemble human interaction. This feasibility study does not consider deliberate spoofing or replay of ultrasonic sensing signals.

Defender capability: We assume that the defender can request browser permission to access the user’s built-in speakers and microphone. EchoGuard emits predefined ultrasonic signals and captures their reflections to study whether browser events are accompanied by physical human interaction. Audible-frequency components are filtered locally before sensing data is transmitted to the server, reducing privacy exposure.

IV. SYSTEM DESIGN

Our proposed system EchoGuard is a browser-based client-server prototype for studying continuous human-presence verification using active ultrasound. As shown in Figure 1, the browser emits a dual-band triangular FMCW signal through the laptop speakers and records its reflections with the microphone. The server extracts acoustic features and aligns them with browser input events to examine whether the inputs are accompanied by corresponding physical activity.

A. Sensing Signal Design

To characterize acoustic responses associated with human and automated-agent interactions, we designed a sensing waveform for commodity laptops using only their built-in stereo speakers and microphone. In this case, a key challenge is that speaker and microphone placement varies across laptops (Figure 2), causing the same keyboard or touchpad movement to produce different reflection strengths along different acoustic

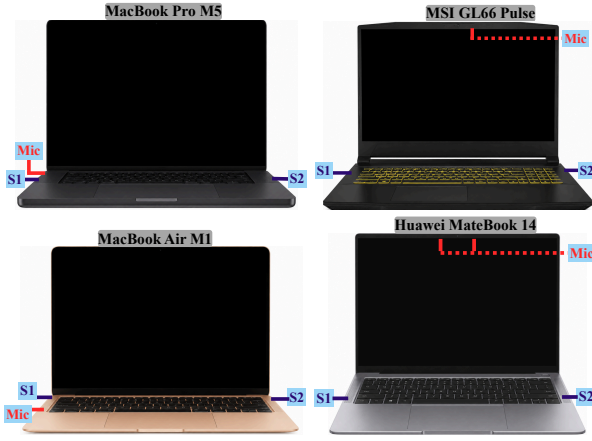


Fig. 2: The built-in speaker and microphone layouts of the four evaluated laptops. S1 and S2 denote the two speakers.

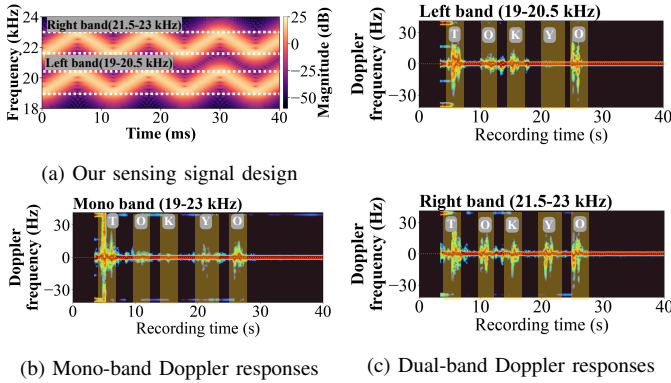


Fig. 3: (a) Our dual-band triangular FMCW signal design provides complementary observations of the same human-typed keyboard sequence T-O-K-Y-O. Compared with (b) mono-band sensing, (c) an event with a weak Doppler response in the left band can remain visible in the right band.

paths. As shown in Figure 3(a), to improve robustness to this variation, we design a dual-band triangular FMCW waveform that provides two complementary observations of the same physical interaction. Each chirp lasts for 12 ms, containing 576 samples at 48 kHz. The left speaker transmits the left band, which sweeps from 19.0 to 20.5 kHz, while the right speaker transmits the right band, which sweeps from 21.5 to 23.0 kHz. The two bands are separated by a 1 kHz guard band. Each band sweeps upward for 6 ms and downward for 6 ms, and the waveform repeats continuously without phase discontinuities at chirp boundaries.

The dual-band design reduces dependence on any single speaker-microphone path by providing two complementary observations of the same interaction. To illustrate this benefit, we compare mono-band and dual-band sensing on the MacBook Pro M5 laptop using the same human-typed sequence T-O-K-Y-O. As shown in Figure 3(b), mono-band sensing uses only the left speaker to transmit a single 19–23 kHz band, producing one Doppler response for the interaction. In contrast, Figure 3(c) uses both speakers to transmit separate left and right bands, producing two speaker-specific Doppler

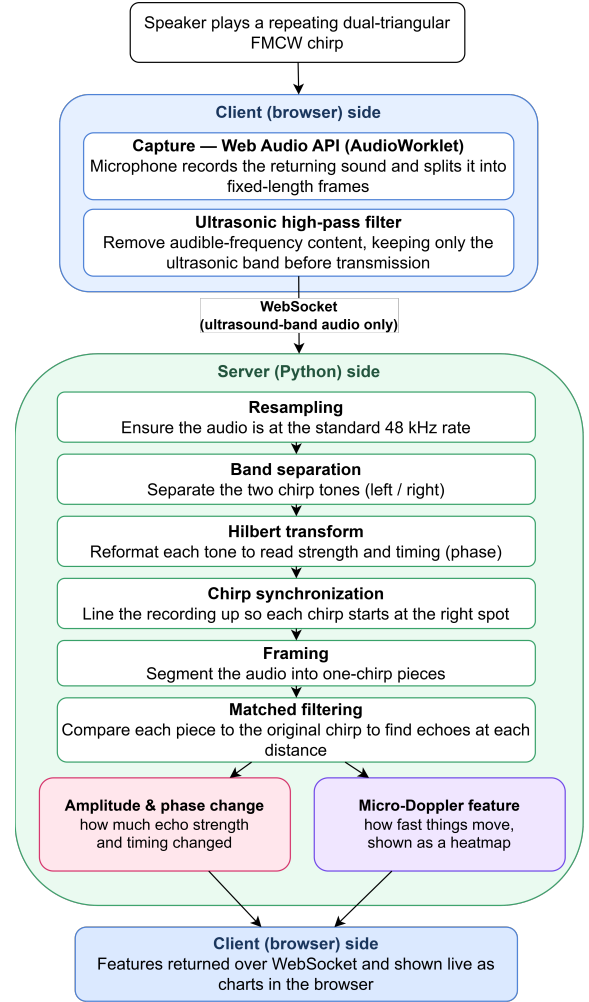


Fig. 4: EchoGuard sensing pipeline. The client captures and filters ultrasonic FMCW reflections, while the server extracts amplitude, phase, and micro-Doppler features for analysis.

responses. Because the two acoustic paths differ, the same keystroke can appear weak in the left-band Doppler response but remain clearly visible in the right-band response. This comparison shows that dual-band sensing introduces path diversity and makes interaction-related Doppler responses more consistently observable.

B. EchoGuard Client-Server Architecture

As shown in Figure 4, EchoGuard divides the sensing pipeline between a browser client and a server. The client handles device-facing and privacy-sensitive operations, while only filtered ultrasonic signals are transmitted over the *WebSocket*. The server performs signal processing and returns sensing features for live visualization and analysis.

Client side. When a sensing session begins, the browser repeatedly transmits a dual-band triangular FMCW chirp through the laptop’s built-in speakers while simultaneously recording the reflected acoustic signal with the microphone. EchoGuard uses the Web Audio API to coordinate real-time playback and capture. An *AudioWorklet* runs on the

browser’s audio-processing thread and segments the incoming microphone stream into fixed-length frames, providing stable timing for subsequent signal processing. Before any audio leaves the device, the client applies a high-pass filter to remove audible-frequency components and retain only the ultrasonic sensing band, thereby preventing intelligible speech from being transmitted. The filtered frames are then streamed to the server over a persistent *WebSocket* connection for low-latency processing. In parallel, the client records timestamped keyboard, pointer movement, scrolling, and clicking events so that the returned acoustic features can be temporally aligned with the corresponding browser interactions. The client also displays the processed sensing features in real time and, at the end of a session, exports the ultrasonic recording, event logs, metadata, and visualization snapshots for offline analysis.

Server side. The Python server hosts the web application and provides a *WebSocket* endpoint for real-time signal processing. As ultrasonic frames arrive, the server maintains processing state across batches and resamples the audio to 48 kHz when needed. We follow standard acoustic sensing procedures to derive amplitude, phase, and Doppler features from the reflected signals [24]. The server separates the two sensing bands using causal band-pass filters and constructs their complex analytic signals via the Hilbert transform. After synchronizing the received signal with the transmitted waveform, the server segments it into individual chirp periods and applies normalized matched filtering to estimate a complex range profile for each chirp. Consecutive range profiles are then compared to derive amplitude and phase changes [29]. In parallel, the server computes micro-Doppler features [30] by applying a 256-point FFT along the slow-time dimension of a sliding 64-chirp (0.768 s) Hann window, advanced by eight chirps (0.096 s) per update. This produces a time–Doppler representation of motion in both ultrasonic bands. All processing is causal and uses only current and previously received samples. The extracted features are returned through the same *WebSocket* for live visualization and subsequent analysis.

V. EXPERIMENTAL SETUP AND EVALUATION

A. Experimental Setup

We evaluate EchoGuard on four commodity laptops with different built-in speaker–microphone configurations: an Apple MacBook Pro M5, a MacBook Air M1, an MSI Pulse GL66, and a Huawei MateBook 14. The devices span both macOS and Windows platforms and differ in the relative placement of their built-in speakers and microphones. No external acoustic hardware is used. On each laptop, the browser continuously transmits the configured FMCW sensing waveform through the built-in speakers while simultaneously recording the reflected signal with the built-in microphone. In parallel, the browser logs timestamped keystroke, pointer-movement, scrolling, and clicking events. These timestamps are used to align each browser interaction with its corresponding acoustic response.

We collect 40-second recordings covering seven different activities: touchpad, clicking, pointer movement, scrolling, sitting still, hand waving, and walking. For human trials,

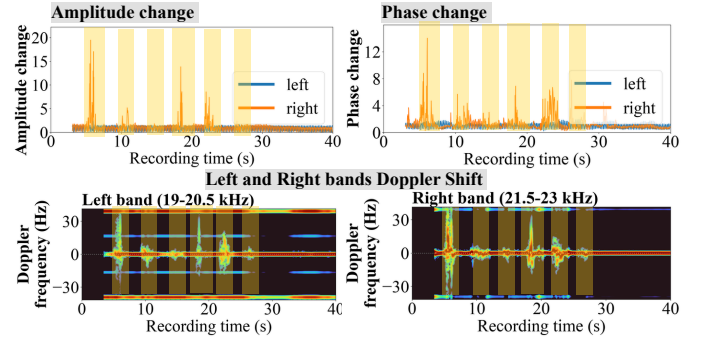


Fig. 5: Comparison of amplitude, phase, and Doppler responses during a touchpad trial. The six shaded intervals denote, from left to right, scroll, move, click, scroll, move, and click events. Doppler responses show the clearest event-aligned motion patterns across the two sensing bands.

two participants perform interactions directly on four different laptops, producing 82 sensing recordings in total. For agent trials, the same types of browser events are generated entirely in software while the participant remains still, producing 10 sensing recordings in total. This allows us to examine the acoustic response when digital input is present without corresponding physical interaction. Each trial begins and ends with a 5-second still period to provide a baseline, with predefined actions performed at fixed intervals during the middle of the recording. We timestamp and label all browser events as human- or agent-generated. To support controlled comparison across trials and devices, we keep the room setup, laptop placement, participant position, speaker volume, sensing parameters, recording duration, and action timing fixed.

B. Feasibility Findings

We analyze the collected data to understand which acoustic features are most informative for capturing physical interaction and how these responses vary across activities and sensing conditions. Our evaluation yields four main findings.

Finding 1: Doppler provides the clearest event-aligned representation among the evaluated acoustic features. Figure 5 compares amplitude change, phase change, and Doppler responses from the same representative touchpad recording on the MacBook Pro M5. These three features capture different aspects of the reflected acoustic signal. Amplitude change reflects variations in echo strength caused by changes in the propagation path, while phase change captures small variations in path length and timing. Both features respond to touchpad interaction, but their visibility varies across actions and sensing bands. Some events produce clear changes, whereas others appear weak or diffuse, likely due to differences in speaker–microphone geometry, hardware response, and the relative position of the user’s hand. In contrast, the Doppler representation directly captures motion-induced frequency shifts over time. Touchpad actions such as scrolling, pointer movement, and clicking produce localized Doppler activity around the corresponding event timestamps, making the temporal relationship between physical motion and browser input easier to

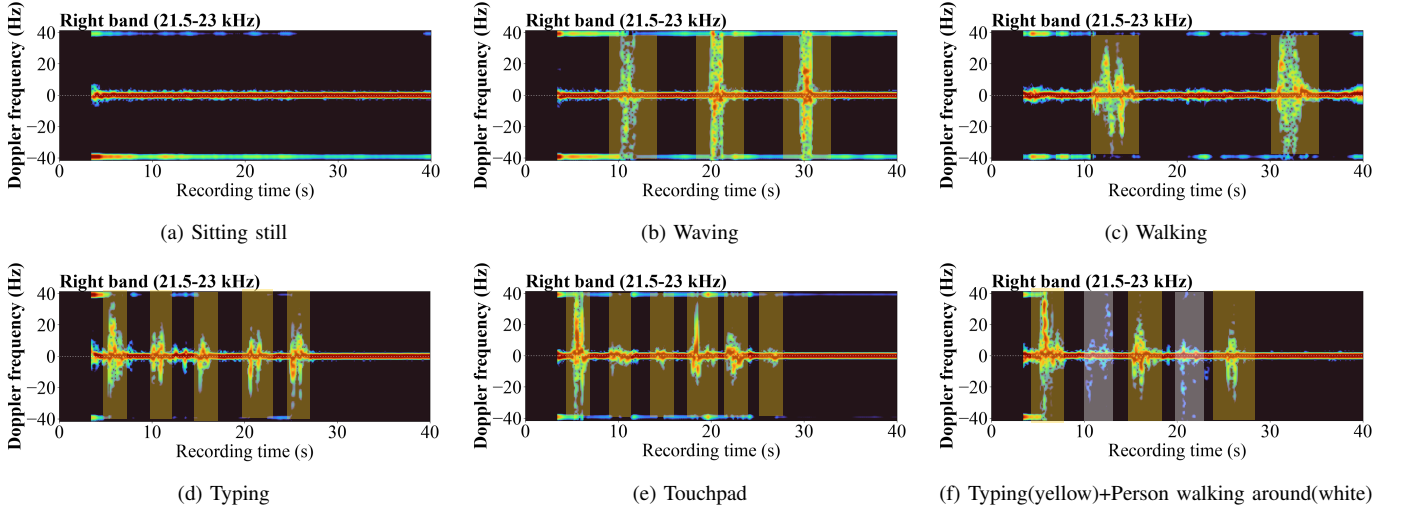


Fig. 6: Doppler responses across six activity conditions in the right sensing band. Shaded regions indicate the corresponding activity intervals. The typing sequence is T-O-K-Y-O, and the touchpad sequence is scroll, move, click, scroll, move, click. In (f), yellow marks typing and white marks background walking.

observe. The strength of the response still varies between the two sensing bands, indicating that Doppler remains dependent on the underlying acoustic path. However, at least one band typically provides a visible response for the evaluated interactions. Among the three evaluated representations, Doppler provides the most consistent and interpretable event-aligned indication of physical interaction in our recordings. We therefore use Doppler as the primary sensing representation for the remaining feasibility analysis.

Finding 2: Direct interaction produces distinct Doppler patterns from surrounding motion. Figure 6 compares Doppler responses across six conditions, including a static baseline, non-interaction motion, direct laptop interaction, and typing under background-motion interference from a single sensing recording on the MacBook Pro M5. Sitting still produces little variation, waving generates intermittent responses, and walking causes broader Doppler activity as the person moves relative to the laptop. These results show that nearby human motion can introduce interference even without direct device interaction. In contrast, typing and touchpad activities produce responses that are temporally aligned with the logged interaction intervals. Typing generates repeated Doppler responses from finger and hand motion, while touchpad use produces more localized and generally weaker responses associated with scrolling, pointer movement, and clicking. When another person walks in the background during typing, both interaction-aligned motion and background interference appear in the Doppler representation. The background response is weaker than in the standalone walking condition because the seated user partially blocks the acoustic path. Overall, the observed Doppler features depend on both the interaction itself and surrounding human motion.

To quantify these differences, we analyze 45 sensing recordings using Doppler bandwidth and Doppler peak, with nine recordings per activity across four laptops. Doppler bandwidth measures the spread of motion-related energy, while Doppler

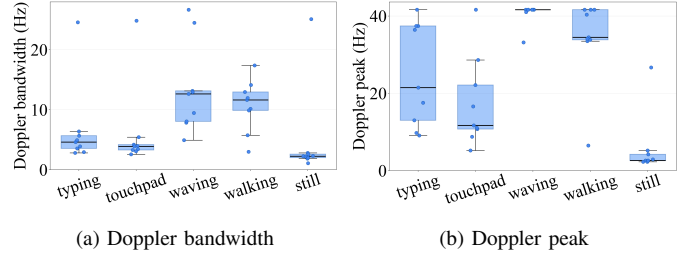


Fig. 7: Doppler statistics across five activity conditions. (a) Bandwidth of motion-related energy and (b) maximum absolute Doppler frequency above the noise floor.

peak captures the maximum absolute Doppler frequency above the noise floor. Figure 7 shows that the five activities occupy different motion regimes. Still recordings remain concentrated at low Doppler frequencies, whereas typing and touchpad interactions exhibit smaller bandwidths and peaks, consistent with localized finger and hand motion. In contrast, waving and walking involve larger body movements and produce broader Doppler distributions and larger peak excursions. These statistics further characterize the difference between direct interaction and larger surrounding-body motion.

Finding 3: EchoGuard captures consistent interaction-aligned Doppler responses across different laptop configurations. We evaluate EchoGuard on four laptops with different built-in speaker-microphone layouts during typing and touchpad interactions. Figure 8 compares Doppler responses from two sensing recordings collected from the same participant using a MacBook Pro M5 and an MSI Pulse GL66, which represent different speaker-microphone layout configurations. Although the Doppler magnitude and relative response strength vary across laptops due to differences in speaker-microphone placement, ultrasonic frequency response, hardware gain, and propagation paths, both devices exhibit event-aligned responses for typing and touchpad actions. These re-

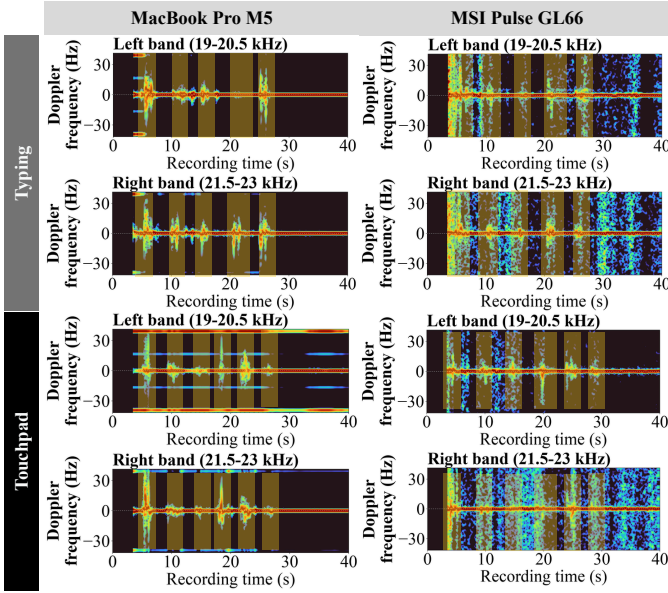


Fig. 8: Typing and touchpad Doppler responses on the MacBook Pro M5 and MSI Pulse GL66. Shaded regions mark the human-generated interaction intervals: T-O-K-Y-O for typing and scroll, move, click, scroll, move, click for touchpad use.

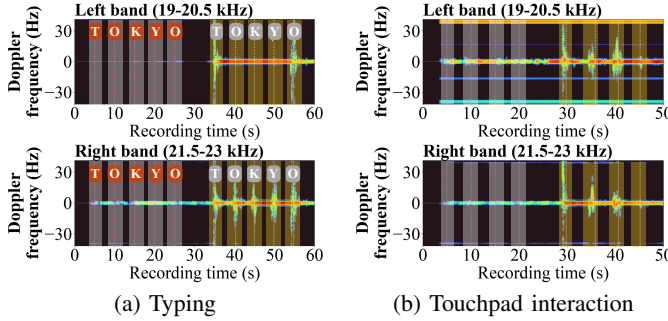


Fig. 9: Doppler responses for human- and AI-agent-generated typing and touchpad interactions. White regions indicate agent events with little physical response; yellow regions indicate human actions with event-aligned Doppler activity.

sults do not imply that Doppler is invariant to laptop geometry; rather, they show that the same sensing and processing pipeline can capture interaction-related motion across laptop setups.

Finding 4: Event-level Doppler features reveal differences between human- and agent-generated interactions across laptops. Figure 9 compares Doppler responses of human and agent typing and touchpad interactions from a single MacBook Pro M5 recording. Human actions produce event-aligned Doppler activity from physical hand and finger motion, while software-generated agent events show little corresponding response. We further analyze all collected data by representing each timestamped event with a 0.5 s window (0.1 s before to 0.4 s after) and extracting a 10-dimensional feature vector comprising micro-Doppler energy, Doppler spread, mean absolute Doppler frequency, signed Doppler centroid, and active duration from each sensing band. We stan-

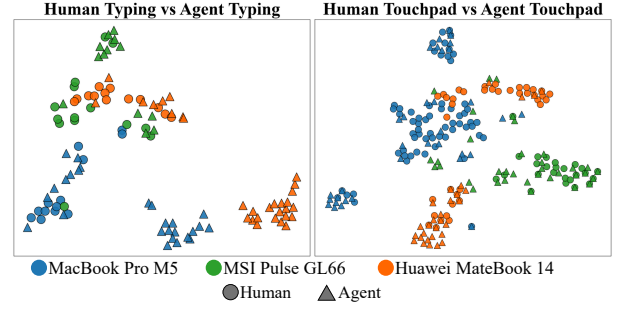


Fig. 10: Two-dimensional t-SNE projections of event-level Doppler features for typing and touchpad interactions across three laptops. Each point represents one event; colors denote laptops, while circles and triangles denote human- and agent-generated events, respectively.

TABLE I: Human-agent typing event classification results.

Model	Accuracy (%) $\uparrow$	Macro F1 $\uparrow$	AUC $\uparrow$
Logistic Regression	89.6	0.793	0.759
Random Forest	87.0	0.720	0.757
CNN	89.6	0.793	0.749

dardize these features and project them into two dimensions using t-SNE [31] (Figure 10). For both typing and touchpad interactions, human- and agent-generated events show distinct distributions across all three laptops, providing initial evidence that event-level Doppler features capture physical differences that remain observable across laptop configurations.

To further validate whether these differences support automated classification, we train three binary classifiers—logistic regression, random forest, and a convolutional neural network (CNN)—to distinguish human from agent-generated typing. Using 16 human and 10 agent recordings, we discard the first and last 5 s and extract 0.25 s event windows, yielding 80 human and 96 agent samples. Logistic regression and CNN achieve 89.6% accuracy and a macro F1 score of 0.793, while random forest achieves 87.0% accuracy and 0.720 macro F1 (Table I). AUC values of 0.749–0.759 further confirm the models’ discriminative ability. Together, they show that we can distinguish human from agent-generated typing events.

VI. CONCLUSION AND FUTURE WORK

EchoGuard explores active ultrasound for continuous human-presence verification against AI agents. Using only built-in laptop speakers and microphones, our feasibility study shows that event-aligned Doppler responses distinguish physical characteristics of human and software-generated keyboard and touchpad interactions across laptop configurations. Future work will expand the dataset, develop an end-to-end classifier for near-real-time verification, and evaluate robustness against replay attacks, environmental variation, and adaptive adversaries. These efforts will assess the practicality of ultrasound-based human-presence verification in real-world web settings.

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